STRONG TIME COMPACTNESS: THREE ROUTES TO ONE CRITERION

Spatial estimates may control u while an evolution differentiates $K(u)$. On a cone, the gauge may not apply to differences. Which spatial and time estimates still force strong compactness?

All three routes produce compact interval integrals. Linear averaging is direct; nonlinear data need truncation. Cone data also need a two-point norm upgrade.

All spaces and the finite interval $(0,T)$ are fixed. Relative compactness means that every sequence has a norm-convergent subsequence. Write

$$\tau_h f(t) = f(t+h), \qquad 0 < t < T-h.$$

Every translation limit below is uniform over the stated family. At $p=\infty$, an L^∞ class needs a continuous representative before it can be treated in $C([0,T];E)$.

We assume Bochner integration, Hölder and Minkowski inequalities, completeness, and the fundamental theorem for $W^{1,1}$ trajectories. The compactness criterion is imported precisely; the nonlinear constructions are developed here.

1. Why interval integrals organize the linear route

Theorem 1.1 (Simon)**.** *Let $T>0$, $1\le p\le\infty$, and let X,B,Y be Banach spaces with continuous linear injections*

$$X \Subset B \hookrightarrow Y,$$

where the first injection is compact. Let $F\subset L^p(0,T;X)$ satisfy

$$\sup_{f\in F}\|f\|_{L^p(0,T;X)}<\infty, \qquad \lim_{h\downarrow 0}\sup_{f\in F}\|f(\cdot+h)-f\|_{L^p(0,T-h;Y)}=0.$$

Then F is relatively compact in $L^p(0,T;B)$ for $p<\infty$. For $p=\infty$, the B-valued continuous representatives of the elements of F form a relatively compact set in $C([0,T];B)$.

This representative statement is [3, Theorem 5, pp. 84–85]. Its spatial and temporal estimates live in different spaces. The next criterion isolates the object that transfers between them.

Lemma 1.2 (Simon)**.** *Let E be a Banach space, $T>0$, $1\le p\le\infty$, and $F\subset L^p(0,T;E)$. Assume that for every $0<a<b<T$ the set*

$$\left\{\int_a^b f(t)\,dt \,:\, f\in F\right\}$$

is relatively compact in E, and that

$$\lim_{h\downarrow 0}\sup_{f\in F}\|f(\cdot+h)-f\|_{L^p(0,T-h;E)}=0.$$

Then F is relatively compact in $L^p(0,T;E)$ if $p<\infty$. If $p=\infty$, its elements have continuous representatives on $[0,T]$ and these representatives form a relatively compact subset of $C([0,T];E)$ with the supremum norm.

This is the sufficient direction of [3, Theorem 1, pp. 71–73]. Its proof uses right time averages and then reverses time to cover the other endpoint.

It reduces each route to two joint tasks: compact interval integrals and small translations in the same ambient norm. At infinity it also supplies continuous representatives; it does not characterize all compact subsets of L^∞.

Put $M_X = \sup_{f\in F} \|f\|_{L^p(X)}$. For each interior interval I, Hölder gives

$$\left\| \int_I f(t)\,dt \right\|_X \le |I|^{1-1/p} M_X.$$

These integrals are relatively compact in B and hence in Y. Lemma 1.2, applied in Y, gives an $L^p(Y)$-Cauchy subsequence.

The remaining operation changes norms. Compactness and injectivity yield the Ehrling estimate [3, Lemma 8, pp. 84–85]:

$$\|z\|_B \le \eta\|z\|_X + C_\eta\|z\|_Y, \qquad z \in X, \quad \eta > 0. \tag{1.1}$$

Indeed, failure at η_0 gives a sequence normalized to unit X norm, with B norm above η_0 and Y norm tending to zero. Compactness gives a B limit; injectivity makes it zero, a contradiction.

Apply (1.1) to $f_n(t) - f_m(t)$ and use Minkowski:

$$\|f_n - f_m\|_{L^p(B)} \le 2\eta M_X + C_\eta\|f_n - f_m\|_{L^p(Y)}. \tag{1.2}$$

For $M_X > 0$, choose $\eta < \varepsilon/(4M_X)$ first. Then choose n, m to make the second term smaller than $\varepsilon/2$. If $M_X = 0$, the family vanishes.

Completeness gives the $L^p(B)$ limit. At infinity, on the full-measure dense set where $\|f(t)\|_X \le M_X$, the same estimate gives uniform B continuity from the continuous Y representative.

First choose η, then $|t - s|$; extend to $[0, T]$ by completeness. The injection identifies the extension with the Y representative. Thus the supremum conclusion in Theorem 1.1 concerns actual continuous functions.

2. How nonlinear images retain compact averages

Theorem 2.1 (Maitre). *Let $T > 0$, $1 \le p \le \infty$, and let E_1, E_2 be Banach spaces. Let $K : E_1 \to E_2$ map every bounded subset of E_1 into a relatively compact subset of E_2. Let $U \subset L^1(0, T; E_1)$ be bounded and suppose that each $v(t) = K(u(t))$, $u \in U$, is strongly measurable. Write $V = \{K(u) : u \in U\}$ and assume $V \subset L^p(0, T; E_2)$, that for some $1 < r \le \infty$*

$$\sup_{v\in V} \|v\|_{L^r(0,T;E_2)} < \infty, \qquad \lim_{h\downarrow 0} \sup_{v\in V} \|v(\cdot + h) - v\|_{L^p(0,T-h;E_2)} = 0.$$

Then V is relatively compact in $L^p(0, T; E_2)$ for $p < \infty$. For $p = \infty$, the continuous representatives form a relatively compact set in $C([0, T]; E_2)$. No injectivity, linearity or monotonicity of K is required. The conclusion concerns $K(u)$, not u.

Maitre's theorem [2, Theorem 2.1, p. 1726] changes the compact unknown to $v = K(u)$. Averaging u cannot be pushed through an arbitrary K. Truncation solves the spatial problem before any averaging occurs.

Let $C_U = \sup_{u\in U} \|u\|_{L^1(E_1)}$ and $C_V = \sup_{v\in V} \|v\|_{L^r(E_2)}$. For $M > 0$, define

$$G_M(u) = \{t : \|u(t)\|_{E_1} > M\}, \qquad u_M = u\mathbf{1}_{G_M(u)^c}, \qquad v_M = K(u_M).$$

Markov bounds the exceptional set. Since v_M equals either v or the fixed value $K(0)$, it is measurable and

$$|G_M(u)| \le C_U/M, \qquad v - v_M = \mathbf{1}_{G_M(u)}(v - K(0)).$$

Hölder therefore gives the uniform tail estimate

$$\|v - v_M\|_{L^1(E_2)} \le (C_U/M)^{1-1/r} C_V + (C_U/M)\|K(0)\|_{E_2}. \tag{2.1}$$

The term $K(0)$ cannot be dropped for an arbitrary compact map. The right side tends to zero for $r > 1$, with $1/r = 0$ at $r = \infty$.

Only an L^1 error is needed: it controls the error of every interval integral. We do not infer a small L^p tail when $p > r$.

For fixed M, all values of v_M lie in the compact set

$$Q_M = \overline{K(\{x : \|x\|_{E_1} \le M\})}.$$

To average such values, take a finite δ-net $q_1, \dots, q_J$ of a compact set $Q \subset E$. Replacing each point of a convex combination by a nearby net point shows

$$\operatorname{co}(Q) \subset \operatorname{co}\{q_1, \dots, q_J\} + \delta B_E.$$

The finite-dimensional simplex is compact. Thus the hull is totally bounded, and its closure is compact by completeness.

For measurable $w : I \to Q$, use the first nearest net point to break ties. The resulting cells are measurable. Their relative measures are nonnegative weights summing to one, so

$$|I|^{-1} \int_I w(t)\, dt \in \overline{\operatorname{co}}(Q).$$

Indeed, the simple-function average lies in the finite simplex and its error is at most δ. Letting δ decrease gives the displayed inclusion.

Hence $\{\int_I v_M\}$ is relatively compact. Choose M to control (2.1), then a finite net for this fixed truncated set. The original interval integrals are totally bounded in E_2.

Lemma 1.2 now uses the assumed $L^p(E_2)$ translations to conclude Theorem 2.1. Maitre uses shifted Riemann sums [2, Lemma 2.3, pp. 1726–1730]; the compact-hull proof here is a local reconstruction.

At $r = 1$, the power $1 - 1/r$ in (2.1) is zero. This bound no longer proves uniform smallness of the discarded output. It does not prove that all nonlinear compactness conclusions fail at $r = 1$.

The account is restricted to these three source interfaces. Later nonlinear translation refinements are outside its scope; no optimality or strongest-known claim is made for the auxiliary $r > 1$ premise.

3. How a cone combines truncation with a norm upgrade

Theorem 3.1 (Barrett–Süli). *Let $T > 0$, let A_0, A_1 be real Banach spaces with a continuous linear injection $A_0 \hookrightarrow A_1$, and let $\mathcal{M}_+ \subset A_0$ be closed under multiplication by nonnegative scalars. Suppose its gauge $[\cdot] : \mathcal{M}_+ \to [0,\infty)$ satisfies*

$$[z] = 0 \iff z = 0, \qquad [cz] = c[z] \quad (c \geq 0), \qquad \|z\|_{A_0} \leq C_0[z],$$

and every gauge-bounded sequence in $\mathcal{M}_+$ has a subsequence converging in A_0. Fix $1 \leq p, q \leq \infty$ with either $q > 1$, or $q = 1$ and $p < \infty$. Let F be a family of strongly measurable A_0-valued functions v on $(0,T)$ with $v(t) \in \mathcal{M}_+$ almost everywhere and measurable $t \mapsto [v(t)]$. Assume the distributional derivatives v' are A_1-valued and

$$\sup_{v \in F} \left(\| [v(\cdot)] \|_{L^p(0,T)} + \|v'\|_{L^q(0,T;A_1)} \right) < \infty.$$

Then F is relatively compact in $L^p(0,T;A_0)$. The assertion includes $p = \infty$ only when $q > 1$; it does not assert that limits belong to a nonclosed cone.

The statement is [1, Theorem 1, pp. 3–4]. The cone has no subtraction operation, while derivatives are measured in A_1. The route must solve both problems: keep gauges on separate points and build time compactness in the weaker space.

Lemma 3.2 (Barrett–Süli). *Let $A_0 \hookrightarrow A_1$ be a continuous linear injection of real Banach spaces. Let $\mathcal{M}_+ \subset A_0$ be a seminormed nonnegative cone with gauge $[\cdot]$, satisfying $\|z\|_{A_0} \leq C[z]$ and compactness of gauge-bounded sequences in A_0. Then for every $\eta > 0$ there is $C_\eta < \infty$ such that*

$$\|u - v\|_{A_0} \leq \eta([u] + [v]) + C_\eta \|u - v\|_{A_1} \quad \text{for all } u, v \in \mathcal{M}_+.$$

Here the cone is closed under nonnegative scalar multiplication and its gauge is positive definite and positively degree-one homogeneous; no addition or subtraction closure is assumed.

The proof in [1, Lemma 1, pp. 4–5] explains why the sum of two gauges appears. If the estimate fails at $\eta_0 > 0$, choose u_n, v_n with

$$\|u_n - v_n\|_{A_0} > \eta_0([u_n] + [v_n]) + n\|u_n - v_n\|_{A_1}.$$

The denominator $a_n = [u_n] + [v_n]$ is positive. Set $\widetilde{u}_n = u_n/a_n$ and $\widetilde{v}_n = v_n/a_n$. Each normalized gauge is at most one.

A common subsequence converges in A_0. Its difference is bounded there, while

$$\|\widetilde{u}_n - \widetilde{v}_n\|_{A_0} > \eta_0, \qquad \|\widetilde{u}_n - \widetilde{v}_n\|_{A_1} < n^{-1}\|\widetilde{u}_n - \widetilde{v}_n\|_{A_0} \longrightarrow 0.$$

Continuity and injectivity force the two A_0 limits to agree, contradicting the separation. Scaling each point stays inside the cone; taking a gauge of their difference would be illegitimate.

3.1. Time control in the space where the derivative is known.

For $v \in W^{1,1}(0,T;E)$, choose a time with norm no larger than the time average plus an arbitrary error. Integrate v' from that time and let the error vanish:

$$\|v\|_{L^\infty(E)} \leq T^{-1}\|v\|_{L^1(E)} + \|v'\|_{L^1(E)}. \tag{3.1}$$

If the resulting amplitude bound is K_0 and $\|v'\|_{L^1(E)} \le D$, Fubini gives

$$\|\tau_h v - v\|_{L^1(E)} \le \int_0^{T-h} \int_t^{t+h} \|v'(s)\|_E \, ds \, dt \le hD. \tag{3.2}$$

The pointwise difference is at most $2K_0$. Hence, for every finite $p \ge 1$,

$$\|\tau_h v - v\|_{L^p(E)}^p \le (2K_0)^{p-1} \|\tau_h v - v\|_{L^1(E)} \le (2K_0)^{p-1} hD. \tag{3.3}$$

For $q > 1$, Hölder on $(t, t+h)$ instead supplies a supremum modulus:

$$\|v(t+h) - v(t)\|_E \le h^{1-1/q} \|v'\|_{L^q(E)}. \tag{3.4}$$

Take $1/q = 0$ for $q = \infty$. At $q = 1$, this last power vanishes. The finite-p estimate has order $h^{1/p}$, not h; compare [3, Lemma 4, pp. 77–78] and [2, Corollary 2.4, p. 1730].

3.2. **Truncate, apply the criterion, then upgrade.** Write $G = \sup_{v \in F} \|[v]\|_{L^p}$, $D = \sup_{v \in F} \|v'\|_{L^q(A_1)}$, and $C_1 = \|A_0 \hookrightarrow A_1\|$. The gauge controls the weak-space time average:

$$\|v\|_{L^1(A_1)} \le C_1 C_0 T^{1-1/p} G.$$

Equation (3.1) then gives

$$\|v\|_{L^\infty(A_1)} \le K_1 := C_1 C_0 T^{-1/p} G + T^{1-1/q} D. \tag{3.5}$$

This supplies continuous A_1 representatives. It does not require cone membership at every time or closedness of the cone.

Cut off by the gauge: $E_R(v) = \{t : [v(t)] > R\}$ and $v_R = v \mathbf{1}_{E_R(v)^c}$. Markov and (3.5) give

$$|E_R(v)| \le \frac{T^{1-1/p} G}{R}, \qquad \|v - v_R\|_{L^1(A_1)} \le \frac{K_1 T^{1-1/p} G}{R}. \tag{3.6}$$

The truncated values have a compact closure in A_0 and hence A_1. The compact-hull argument already proved gives compact interval integrals in A_1; the uniform tail passes this to v.

If $q = 1$ and $p < \infty$, use (3.3) in A_1. If $q > 1$, use (3.4). Lemma 1.2 gives a subsequence Cauchy in $L^p(A_1)$ in every allowed case.

Finally, Lemma 3.2 and Minkowski yield

$$\|v_n - v_m\|_{L^p(A_0)} \le 2\eta G + C_\eta \|v_n - v_m\|_{L^p(A_1)}. \tag{3.7}$$

For $G > 0$, choose $\eta < \varepsilon/(4G)$ before the indices. Completeness gives strong $L^p(A_0)$ convergence. The case $G = 0$ is immediate because every trajectory vanishes almost everywhere.

At $p = \infty$, the full-measure dense set with $[v(t)] \le G$ also gives continuous A_0 representatives. Apply two-point Ehrling to times s, t: choose η, then $|t - s|$, and extend by completeness.

The upgrade agrees with [1, equation (2.22), pp. 12–13]. The weak-space argument here avoids the unproved time-dependent-index step following its (2.8), p. 6. This scoped concern is not a claim that the theorem is false.

4. Synthesis on matching hypotheses

The common criterion receives different inputs. The compact unknown and the space of time control must be retained when comparing the routes.

Route	Spatial operation	Time control and final norm
Linear	Integrate a bounded $L^p(X)$ family	$L^p(Y)$ translations; upgrade to B
Compact map	Truncate u, then average the compact image	$L^p(E_2)$ translations of $K(u)$; limit in E_2
Cone	Truncate by gauge; compare separate points	Derivative in A_1; upgrade to A_0

The cone proof combines operations from the first two routes. This structural similarity does not identify their hypotheses. The following reduction states the extra data needed for a genuine overlap.

Proposition 4.1. *Let E_1, E_2 be real Banach spaces, $T > 0$, and $\alpha \geq 1$. Suppose $K : E_1 \to E_2$ is injective, maps bounded sets into relatively compact sets, and satisfies $K(cu) = c^\alpha K(u)$ for $c \geq 0$. Set $\mathcal{M}_+ = K(E_1)$ and $[v] = \|K^{-1}v\|_{E_1}^\alpha$. Let $1 \leq p, q \leq \infty$, with $q > 1$ or with $q = 1, p < \infty$. Suppose a family F of strongly measurable $\mathcal{M}_+$-valued trajectories has strongly measurable inverse lifts $u = K^{-1}v$, measurable gauges, and*

$$\sup_{v \in F} \left(\| [v] \|_{L^p(0,T)} + \| v' \|_{L^q(0,T;E_2)} \right) < \infty,$$

where derivatives are E_2-valued distributions. Then the hypotheses of the compact nonlinear-map theorem stated for K, $U = K^{-1}F$, and $V = F$ hold, using any fixed $r > 1$ and the target exponent p. Consequently F is relatively compact in $L^p(0,T;E_2)$, and in $C([0,T];E_2)$ when $q > 1$. This reduction concerns the special cone $K(E_1)$ with the same output and derivative space E_2; it is not an equivalence for arbitrary cones or arbitrary compact maps.

The special reduction in [1, Section 4, pp. 15–16] can be checked directly. Injectivity and homogeneity give $K(0) = 0$ and

$$K^{-1}(cv) = c^{1/\alpha} K^{-1} v, \qquad [cv] = c[v].$$

The compact image of the unit ball is bounded. Scaling therefore gives

$$\|K(x)\|_{E_2} \leq C \|x\|_{E_1}^\alpha = C[K(x)].$$

Gauge-bounded sets come from bounded input sets and are relatively compact in E_2. This verifies the cone axioms and compact embedding.

Let $G = \sup \| [v] \|_{L^p}$. Since $\alpha p \geq 1$, Hölder gives

$$\|K^{-1}v\|_{L^1(E_1)} \leq T^{1-1/(\alpha p)} G^{1/\alpha}, \tag{4.1}$$

with factor T at $p = \infty$. The assumed measurable inverse lifts thus form a bounded $L^1(E_1)$ family.

The gauge and derivative bounds give a uniform $W^{1,1}(E_2)$ bound. Equation (3.1) supplies every fixed $L^r(E_2)$ bound, $r > 1$; (3.3) or (3.4) supplies the required translations.

Theorem 2.1 applies, using $p = \infty$ when $q > 1$ to obtain the C conclusion. The derivative must be measured in E_2, the output space. A derivative in a larger weak space cannot simply be substituted.

Nor does an arbitrary cone come with this parametrization. Injectivity is needed for this inverse gauge; it is not an assumption of Maitre's theorem. These are limits of the reduction, not proofs that every alternative construction fails.

4.1. **Two tests of what the comparison cannot imply.** Take $E_1 = E_2 = \mathbb{R}$, $K(x) = x^2$, and

$$u_n(t) = (-1)^{\lfloor 2^n t \rfloor}, \qquad 0 < t < 1.$$

For $m > n$, each dyadic cell for u_n contains an even number of alternating cells for u_m. The signs differ on exactly half the interval, so

$$\|u_n - u_m\|_{L^1} = 2 \cdot \tfrac{1}{2} = 1, \qquad K(u_n) = 1.$$

The inputs have no L^1-Cauchy subsequence, although their images are a singleton with zero translations. This tests the exact unknown in [2, Theorem 2.1 and Remark 2.2(3), p. 1726].

It also prevents using a formal K^{-1} for every compact map. The example contradicts input recovery; it fully agrees with compactness of the images.

On $(0, 1)$, let $n \geq 3$ and set

$$v_n(t) = \begin{cases} 0, & t \leq \frac{1}{2}, \\ n(t - \frac{1}{2}), & \frac{1}{2} < t < \frac{1}{2} + \frac{1}{n}, \\ 1, & t \geq \frac{1}{2} + \frac{1}{n}. \end{cases}$$

These are continuous trajectories in the cone $[0, \infty) \subset \mathbb{R}$, with gauge $[z] = |z|$. Their spatial bounds are harmless, and

$$\|v_n\|_\infty = 1, \qquad \|v_n'\|_1 = n \cdot n^{-1} = 1.$$

Their pointwise limit is the discontinuous step $w = \mathbf{1}_{(1/2,1]}$. No subsequence converges uniformly, since a uniform limit of continuous functions is continuous.

For continuous differences, the supremum and essential supremum agree. An L^∞-Cauchy subsequence would therefore be uniformly Cauchy and have a continuous uniform limit, the same contradiction.

Yet every finite exponent gives convergence:

$$\|v_n - w\|_p^p = \int_0^{1/n} (1 - ns)^p \, ds = \frac{1}{n(p+1)} \longrightarrow 0. \tag{4.2}$$

Thus $(p, q) = (\infty, 1)$ is a genuine obstruction already in one dimension; compare [3, Proposition 3, p. 94] and [1, Remark 2, p. 4].

The synthesis is an interface, not an equivalence. Compact averages and a uniform time modulus are the common outputs of the preliminary arguments; linear structure, compact maps and cone gauges are different ways to produce them.

The inverse-gauge proposition gives a real implication under its stated extra hypotheses. The two examples rule out stronger conclusions for the wrong unknown or exponent. The $r = 1$ tail issue remains a boundary of this selected proof alone.

The controlling texts are Simon's journal pagination and DOI, Maitre's published 2003 article, and Barrett–Süli's arXiv v2 of 16 April 2011. The latter's different internal 2018 date is not used to infer a new version.

The argument imports Simon's criterion and reconstructs the selected mechanisms. It claims no original research result, external referee certification or real-reader study. The historical addition of the cone source is preserved separately from these new readings.

References

[1] John W. Barrett and Endre Suli. Reflections on dubinskii's nonlinear compact embedding theorem. https://arxiv.org/pdf/1101.1990v2, 2011. preprint; controlling version: arXiv:1101.1990v2, submitted 16 April 2011, 17 pages.

[2] Emmanuel Maitre. On a nonlinear compactness lemma in lp(0,t;b). 10.1155/S0161171203106175, 2003. published; controlling version: Published IJMMS 2003:27, pp. 1725–1730, DOI 10.1155/S0161171203106175.

[3] Jacques Simon. Compact sets in the space lp(0,t;b). 10.1007/BF01762360, 1986. published; controlling version: Journal text, Ann. Mat. Pura Appl. 146, pp. 65–96, DOI 10.1007/BF01762360; publisher issue date December 1986 (frequently cited as 1987).